Experiment 4: The Sol-Gel Technique – Lab Report

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1) Aim

To synthesize SiO_2 nanopowder by the sol-gel method (TEOS precursor, NH_4OH/NH_4F catalyst), to produce thin coatings by dip and spin coating, and to examine the effect of SiO_2 content on the wettability (contact angle) and viscosity (Ostwald viscometer) of PVA–SiO_2 suspensions.

2) Materials and Devices

Chemicals: TEOS ($Si(OC_2H_5)_4$), ethanol, distilled water, NH_4F, NH_4OH, PVA, Triton X-100. **Devices:** magnetic stirrer, beakers, dropper, precision balance, fume hood, drying oven, calcination furnace, laser pointer, spin coater, FTO/ITO glass and glass slides, Ostwald viscometer, stopwatch.

3) Experimental Procedure

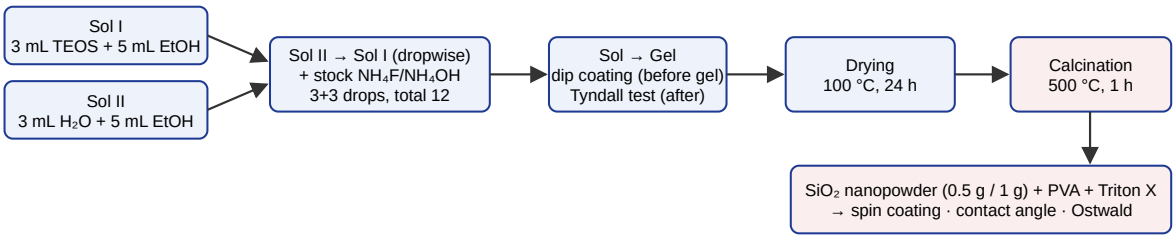


Figure 1. Flow chart of the experiment.

3.1) Sol preparation and gelation

- **Solution I:** 3 mL TEOS + 5 mL ethanol, stirred magnetically in the fume hood. **Solution II:** 3 mL distilled water + 5 mL ethanol.
- **Stock (catalyst) solution:** 1.9 g NH_4F + 23 mL NH_4OH in 100 mL distilled water [1].
- Solution II was added dropwise to Solution I, which started the hydrolysis of the alkoxide: $\equiv Si–OR + H_2O \rightarrow \equiv Si–OH + ROH$, followed by condensation into Si–O–Si bonds [2].
- The stock solution was added in portions of 3 drops with waiting time in between. When a coating started to form on the dipped glass slide, the addition was stopped (12 drops in total). Ethanol is the common solvent for TEOS and water; NH_4OH (base) and F^- accelerate the reaction.

3.2) Dip coating, gel and Tyndall test

Before full gelation, a glass slide was dipped into the sol and slowly withdrawn to obtain a thin silica film. After gelation, a laser beam was passed through a beaker of distilled water and then through the gel.

3.3) Drying and calcination

The wet gel was weighed, dried at 100 °C for 24 h (water and alcohol removed) and calcined at 500 °C for 1 h (residual organics removed), giving the SiO_2 nanopowder. It was weighed after each step.

3.4) Spin coating, contact angle and viscosity

- Two PVA suspensions with 0.5 g (low) and 1 g (high) SiO_2 and Triton X-100 (surfactant) were prepared.
- **Spin coating:** 3 drops on FTO/ITO glass; 500 rpm–5 s, 500 rpm–10 s, 800 rpm–10 s, 800 rpm–5 s.
- **Contact angle:** one drop of distilled water was placed on the high- and low-silica coatings, a graphite-coated paper and a nanofiber paper.
- **Ostwald viscometer:** filled to mark C, the liquid was drawn above A with the bulb and the A→B flow time was measured. Distilled water was the reference; the same volume (5 mL) was used for all liquids and ρ = m/V was found by weighing.

4) Results and Discussion

4.1) Percent weight loss

Step	Mass (g)	Loss (%)
Wet gel	10.900	–
Dried, 100 °C, 24 h	0.576	94.72
Calcined, 500 °C, 1 h	0.394	31.60
Total		**96.39**

Table 1. Mass change of the gel.

$$\textit{Weight loss (\%)} = (m_{before} - m_{after}) / m_{before} \times 100$$

Drying: (10.9 − 0.576)/10.9 × 100 = 94.72 %
Calcination: (0.576 − 0.394)/0.576 × 100 = 31.60 %
Total: (10.9 − 0.394)/10.9 × 100 = 96.39 %

The wet gel is mostly liquid (water, ethanol and the ethanol released by hydrolysis) trapped in the silica network, so most of the mass is lost on drying. The loss during calcination comes from residual $–OC_2H_5$ groups and bound water, and from Si–OH groups

condensing into Si–O–Si. Theoretically, 3 mL TEOS (0.933 g/mL, 208.33 g/mol → 0.0134 mol) gives 0.0134 × 60.08 ≈ 0.81 g SiO_2, so the yield is ≈ 49 %. The rest was used for dip coating or stayed in the beaker and crucible.

4.2) Tyndall effect

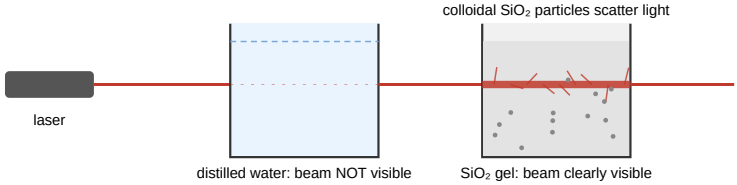
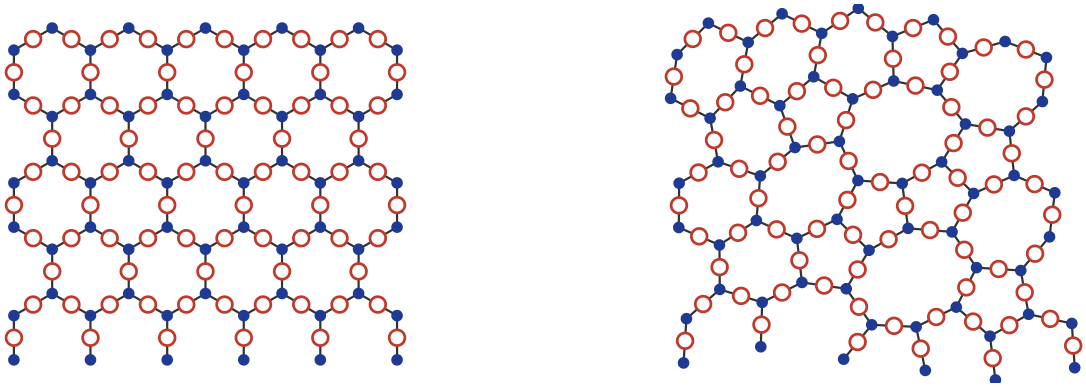


Figure 2. Observation of the Tyndall effect.

The laser beam could not be seen in distilled water but was clearly visible inside the gel. The colloidal SiO_2 particles in the gel are of a size comparable to the wavelength of light and scatter it in all directions, so the path of the beam becomes visible. A true solution has no such particles. This shows that the sol/gel is a colloidal system [2].

4.3) Crystalline and amorphous SiO_2



(a) Crystalline SiO_2 – periodic, regular 6-rings *(b) Amorphous (vitreous) SiO_2 – random network*

● Si ○ O *Figure 3. 2-D schematic of the SiO_4 network (4th O of each tetrahedron out of plane), after Zachariasen [3].*

Property	Crystalline SiO_2 (quartz, cristobalite)	Amorphous SiO_2 (silica glass, sol-gel silica)
Atomic order	Long-range periodic order; regular rings	Only short-range order (SiO_4 tetrahedra); random rings of 4–8 members
Density	Higher (quartz 2.65 g/cm³)	Lower (~2.2 g/cm³; sol-gel powder is porous)
Melting	Sharp melting point (cristobalite ~1713 °C)	No sharp melting point; softens gradually above T_g (~1200 °C)
XRD / optics	Sharp peaks; anisotropic, piezoelectric (quartz)	Broad hump at 2θ ≈ 22°; isotropic
Chemical	Low surface area, few Si–OH, low reactivity	High surface area, many Si–OH groups, more reactive and more soluble

Table 2. Chemical and physical differences between the two structures [2, 3].

Our powder is amorphous. It was formed at room temperature by random hydrolysis and condensation and was only heated to 500 °C. Sol-gel silica stays amorphous at this temperature; crystallization to cristobalite needs much higher temperatures (≳1000 °C) [2]. **Application areas of SiO_2 nanopowder:** filler in polymers and rubber (e.g. tyres), catalyst support and adsorbent (silica gel desiccant), drug delivery (mesoporous silica), anti-reflective, scratch-resistant and superhydrophobic coatings, CMP polishing slurries, anti-caking agent in food and cosmetics, and nano-silica additive in concrete [4].

4.4) Contact angle and superhydrophobic surfaces

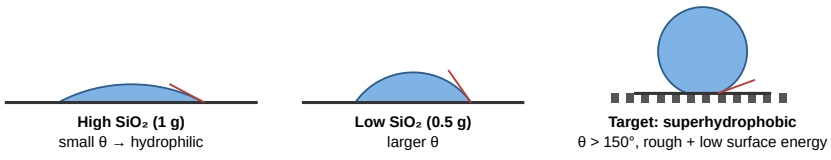


Figure 4. Schematic of the observed droplet shapes and the target surface (angles not measured numerically).

With the Young equation, $\gamma_{SV} = \gamma_{SL} + \gamma_{LV} \cos\theta$, a lower θ means better wetting (θ < 90° hydrophilic, θ > 90° hydrophobic, θ > 150° superhydrophobic) [1]. The drop spread more on the **high-SiO_2** coating than on the low-SiO_2 one, so **the contact angle decreased and the surface became more hydrophilic as SiO_2 increased**. The silica surface is covered with polar Si–OH (silanol) groups, which raise the surface energy and form hydrogen bonds with water. (The nanofiber paper did not absorb the drop and behaved hydrophobically.)

Chemical treatment for a superhydrophobic surface: the silica must be **silanized**, i.e. its Si–OH groups must be replaced by non-polar, low-surface-energy groups [7, 12]:

$$\equiv Si–OH + (C_2H_5O)_3Si–R \rightarrow \equiv Si–O–Si(OC_2H_5)_2–R + C_2H_5OH \quad (R = –CH_2CH_2(CF_2)_nCF_3, \text{ fluoroalkylsilane FAS})$$

$$2 \equiv Si–OH + (CH_3)_3Si–NH–Si(CH_3)_3 \text{ (HMDS)} \rightarrow 2 \equiv Si–O–Si(CH_3)_3 + NH_3$$

Alternatively, part of the TEOS can be replaced by a methyl-silane co-precursor (e.g. MTMS) in the sol, so that Si–CH_3 groups are built into the network [13]. The low surface energy alone gives at most $\theta \approx 120°$ [12]. The SiO_2 nanoparticles add micro/nano roughness (like the lotus leaf [11]); air is trapped under the drop (Cassie–Baxter state [8]) and roughness amplifies the hydrophobicity (Wenzel [9]). The result is $\theta > 150°$ with a sliding angle < 10°, so rain drops roll off and clean the surface.

4.5) Spin coating: effect of rpm and time on thickness

For a Newtonian liquid on a rotating disk, the thickness decreases with time as [5]:

$$h(t) = h_0 / \sqrt{(1 + 4\rho\omega^2 h_0^2 t / 3\eta)}$$

(h: film thickness, h_0: initial thickness, ω: angular speed, ρ: density, η: viscosity, t: time)

Higher rpm → thinner film. Including solvent evaporation, the final thickness follows $h_f \propto \omega^{-1/2}$ [6, 14]; going from 500 to 800 rpm should give $(500/800)^{0.5} \approx 0.79$, i.e. ~20 % thinner film. A higher viscosity (more SiO_2) gives a thicker film. **Longer time → thinner only at the beginning.** Thinning is fast in the first seconds (spin-off), then slows ($h \propto t^{-1/2}$) and stops when evaporation makes the film immobile [6, 14]. Therefore 5 s and 10 s gave practically the same coating in our experiment, and the rpm was the controlling parameter.

4.6) Viscosity of SiO_2-doped PVA solutions

Liquid (V = 5 mL)	m (g)	ρ (g/cm³)	t (s)	η (mPa·s)
Water (ref.)	–	1.000	1.45	0.89
Low SiO_2	5.1293	1.02586	8.06	**5.08**
High SiO_2	5.1332	1.02664	8.34	**5.26**

Table 3. Ostwald viscometer results.

$$\eta = \eta_0 \cdot (\rho \cdot t) / (\rho_0 \cdot t_0)$$

Low: 0.89 × (1.02586 × 8.06)/(1 × 1.45) = 5.08 mPa·s
High: 0.89 × (1.02664 × 8.34)/(1 × 1.45) = 5.26 mPa·s
(η: dynamic viscosity, ρ: density, t: flow time; index 0 = water. Units cancel in the ratio → mPa·s.)

The viscosity increased from 5.08 to 5.26 mPa·s (~3.6 %) as the SiO_2 content increased; both are ~5.7–5.9 times that of water. More solid particles disturb the flow and increase internal friction, and the Si–OH groups on the particles form hydrogen bonds with the –OH groups of the PVA chains. A higher viscosity gives a thicker film in coating. Error sources: the hand-timed and very short reference time (1.45 s), uncontrolled temperature, and using ρ_0 = 1 g/cm³ and η_0 = 0.89 mPa·s instead of the 25 °C values (0.997 g/cm³, 0.8909 mPa·s) [10].

5) Conclusion

Amorphous SiO_2 nanopowder was synthesized by the sol-gel method with a total weight loss of 96.39 %, and the colloidal nature of the gel was shown by the Tyndall effect. Increasing the SiO_2 content made the coating more hydrophilic and increased the viscosity of the PVA solution. A higher spin speed gives a thinner film, while spin time matters only in the first seconds. A superhydrophobic surface requires silica roughness combined with low-surface-energy silanization.

6) References

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